

NARAMREDDY CHARAN KUMAR REDDY

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PROFESSIONAL SUMMARY

Computer Science student specializing in AI and Machine Learning with a focus on building end-to-end data pipelines, predictive modeling, and scalable systems. Experienced in developing recommendation architectures using deep learning and vector index search, and tuning tree-based classifiers using hyperparameter optimization. Strong foundation in Python OOP, SQL, and data separation methodologies for production-oriented machine learning.

TECHNICAL SKILLS

- **Machine Learning:** XGBoost, Random Forest, Logistic Regression, Scikit-Learn, Optuna, Classification Models.
- **Deep Learning & GenAI:** TensorFlow, Keras, Neural Collaborative Filtering, Vector Embeddings, FAISS Similarity Search, Prompt Engineering.
- **Data Intelligence:** Pandas, NumPy, Feature Engineering, KNNImputer, Imbalance Handling, RobustScaler.
- **Engineering Core:** Python 3 (Advanced OOP, Data Structures), SQL (Schema Design, SQLAlchemy), Git Workflow, BeautifulSoup.

ENGINEERING PROJECTS

E-Commerce Customer Churn Prediction | *Python, Scikit-Learn, XGBoost, Optuna* | github.com/NaramCharan/ecommerce-customer-churn-prediction

- Engineered an end-to-end classification pipeline using ColumnTransformer with hybrid KNN/Simple value imputation to clean, scale, and encode customer data while strictly preventing data leakage.
- Tuned tree classifiers using Optuna with 10-fold Stratified Cross-Validation, achieving **98.28% accuracy** and 94.76% F1-score on the final selected XGBoost production model.

Neural Collaborative Filtering Architecture | *TensorFlow, Keras, FAISS, Python* | github.com/NaramCharan/Collaborative_Filtering_Recommendation_system

- Built a custom recommendation engine using TensorFlow and Keras, training 32-dimensional latent embedding vectors on sparse user-item interaction datasets with L2 regularization.
- Integrated Meta's FAISS library to index item embedding vectors, enabling sub-10ms similarity search queries for real-time recommendations.

Titanic Survival Engine: Predictive Analytics | *Python, Scikit-Learn, XGBoost, Optuna* | github.com/NaramCharan/Titanic-Survival-Engine-Predictive-Analysis

- Developed a binary classification pipeline achieving 85.5% accuracy by evaluating XGBoost and Random Forest architectures against a baseline Logistic Regression model.
- Prevented data leakage using custom imputation on training distributions and optimized tree models using Optuna hyperparameter search.

Book Data Scraping & Database Pipeline | *BeautifulSoup, SQLAlchemy, SQLite, Pandas* | github.com/NaramCharan/Book-webscraper

- Built an automated multi-page scraper using BeautifulSoup and SQLAlchemy ORM, extracting structured data (title, price, rating, availability, description, URL) for 980+ books in under 30 minutes.
- Designed a relational SQLite database schema and exported clean CSV datasets ready for downstream machine learning applications, replacing days of manual collection.

EDUCATION

GD Goenka University — Gurugram, India

B.Tech in Computer Science Engineering (AI & Machine Learning)

Graduation: May 2028

Current CGPA: **8.7 / 10.0**

CERTIFICATIONS

- **Machine Learning Specialization** – DeepLearning.AI, Stanford University (May 2026)
- **Databases and SQL for Data Science with Python** – IBM, Coursera (Nov 2025)
- **Python for Everybody Specialization** – University of Michigan, Coursera (March 2025)
- **Prompt Engineering & Generative AI** – Google, Vanderbilt University (March 2025)